APPENDIX

A: Data Extraction Form for Selected Studies

Author(s)	Study Aim(s)	App Intervention (App features)	Participants	Comparator	Study Perspective	Time horizon	Economic method	Outcome(s)	Outcome measure(s)	Data source(s)	Program Effective? How?	Sample Size	Demographics	Behavior targeted	Behavior Theory/construct?	Report Limitations	Report on Participant Engagement	Results (App Valuable)?	Country	Own phone/tablet?	App Free?	Defined outcome?	Well-defined research question?	Described comparator?	All costs of both identified?	Consequences in appropriate physical units?	Costs and Consequences valued credibly?	Time adjustment?	Incremental analysis?	Sensitivity /uncertainty analysis?	Discus cost of/implications for scale?	Addressed potential confounding	Wirefrme/ User interface/Logic diagram?
Van Reijen M, et al. (2017)	To prevent the recurrence of ankle sprains delivered through an App vs the same intervention delivered through a Booklet.	"Strengthen your Ankle" mobile app. Provided written, visual, and verbal instructions and a calendar function.	Active sports participants or athletes (between 18 and 70 years) who had incurred a self-reported ankle sprain within the past 2 months and using the "Strengthen your Ankle" mobile app	Active sports participants or athletes (between 18 and 70 years) using a booklet	Societal	12 months	Cost-effectiveness analysis	Incidence density of ankle injury and incremental cost-effectiveness ratio (ICER)	Incremental cost-effectiveness ratio (ICER), expressed as cost per injury incidence density.	Baseline online questionnaires, use of the injury incidence density (IID) to calculate the number of recurrent ankle sprains per 1000 hours of exposure	Yes, RCT	220	Gender (50% male), age (37.6 +/- 13.1),	Compliance with NMT program.	None	Yes (confounding effects)	Yes	No difference	The Netherlands	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Martín JAC, et al. (2014)	To provide each patient with the necessary technology for the management and monitoring of their disease using an application	Cardiomanager, a medical app to allow heart diseases patients to self-manage their own conditions.	Hypothetical patients with documented chronic HF in the clinical history of primary care	Do nothing	Healthcare sector (Ministry of Health, Social Policy and Equality)	1 year	Cost-effectiveness analysis	Costs savings, care quality, monitoring and patient follow-up, user satisfaction, process efficiency.	Incremental cost-effectiveness ratio (ICER), expressed as cost per additional quality-adjusted life year (QALY) obtained	Costs of the Ministry of Health, Social Policy and Equality	No, ex-ante	630, Hypothetical cohort.	Not stated. Hypothetical cohort.	Adherence to self-management regimen, patient autonomy	None	Yes, ex-ante economic analysis, lack of data; lack of societal perspective	No	Yes, 33% cost savings	Spain	Assumed	Yes	Yes	Yes	No	Yes	Yes	No	Yes, 3%	Not shown	Yes	No	No	No

Bitsaki M, et al. (2017)	To improve the communication between arthroplasty patients and physicians and to reduce the cost of follow-up controls	Combination of cloud- and service-oriented computing, online services, data analysis, and mHealth applications to connect patients to their physicians.	Patients with hip and knee arthroplasty	Usual care	Healthcare sector	1 year	Cost analysis	Cost savings	Cost	questionnaires (WOMAC and SF-36v2); cost estimations based on input taken from primary hip and knee replacement surgeries performed at the University Clinic for Orthopedics in Essen-Werden	NO, based on literature	423	None	Patient-physician communication, patient self-monitoring and regular review of own health status.	None	Yes, assuming patient characteristics are same for scaling	No	Yes, cost savings	Germany	Yes	Yes	Yes	Yes	No	Yes	N/A	Cost, yes	No	Yes	No	Yes	Yes	Yes
Sjöström M, et al. (2017)	To evaluate the health economy of the app for treating Stress Urinary Incontinence	An app Tat is first line treatment for SUI, based on self-management and provides instructions for pelvic floor muscle training (PFMT)	Community-dwelling women of 18 years and above, with stress urinary incontinence ≥1 time per week.	No app treatment	Societal	1 year	Cost-effectiveness analysis/Cost-utility analysis	QALYS gained	ICER, with outcome measured in QALYS	QALYS measured with the International Consultation on Incontinence Modular Questionnaire on Lower Urinary Tract Symptoms and Quality of Life	Yes, RCT	123	Yes, age, education, symptom severity and QoL	Self-management of PFMT	None	Yes	No	Yes, higher utility, cost-effective	Sweden	Assumed	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes, one way and multi way	Yes, additional cost for bug fixes, technical support, updates, etc.	Yes, education	No

Bovbjerg ML, et a. (2017)	To connect high-cost, high-utilizing Medicaid an mHealth App for Remote Case Management of Low-Income Patients with Diabetes to four nurse case managers at a large multispecialty clinic in the Pacific Northwest	An mHealth app for remote case management of low-Income patients with diabetes	Low-Income patients with diabetes	None (pre-post study design)	Not stated	Not specific (different time lengths for different patients)	Cost analysis	Accessibility of clinic staff; provider knowledge; prescription drug behavior; diabetes-specific behaviors and quality of care; lifestyle behavior changes plans; overall patient satisfaction; depressive symptoms; overall quality of life and general health; physical health; psychological health; social relationships; environment; self-efficacy	Cost savings	CAHPS, PHQ-9, WHOQOL-BREF, and Self-Efficacy for Managing Chronic Disease 6-item scale	Yes, pre-post assessment	33 (24)	age, race, income,	Adherence to Self-monitoring and self-management regimens	Self-efficacy	Yes, cannot attribute causality; small sample, pre-post design, inconsistent timeline, multiple comparisons	No	Yes, cost savings	US	Given tablet, with app installed	Yes	Yes	No	N/A	Cost	N/A	Maybe	No	No	No	No	No	No
Dahlberg K, et al. (2017)	To estimate the cost-effectiveness of using RAPP for follow-up on postoperative	Recovery assessment by phone points (RAPP) is a	Patients >17 yr. of age undergoing day surgery	Usual (standard) care	Health-care perspective	2 weeks	Cost-effectiveness analysis	Primary: cost-effectiveness of RAPP use. Secondary: effect of	Health effect [quality-adjusted life-years (QALYs)], and costs or savings in health-care use.	QoL measured with SF-6D; Swedish Web version of Quality of Recovery	Yes, RCT	997 (28 % loss)	Not reported.	Quality of post-operative recovery	None	QALYs not included in CEA, thus performed as	Yes	Yes for cost savings, no difference in QALYS,	Sweden	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes		Yes	No

	recovery compared with standard care.	smartphone-based application (app) to evaluate patients after day surgery						using RAPP on the quality of postoperative recovery.		(SwQoR) questionnaire; yes/no questions. Cost data from business plan, hospital accounting departments; healthcare costs from Swedish patient cost database (KPP data)				through patient engagement.		cost-minimization analysis; loss to follow-up		CE not reportable																
Luxton DD, et al. (2014)	To calculate the cost of providing stress reduction care with the B2R app in comparison with normal office care	Providing stress reduction care with a mobile phone app (Breathe2 Relax) in military and civilian populations through instructions on diaphragmatic breathing techniques.	Military beneficiaries; civilians	Usual (Standard) care	Military healthcare system; civilian healthcare system	1 year	Cost-minimization analysis	Stress reduction through diaphragmatic breathing techniques.	Cost savings	The Current Procedural Terminology (CPT) code used to value the cost of MHS-purchased care; cost of 30 min of patient time was based on 2011regular military compensation estimates;	No, satisfactory theoretical basis	1000	No	Adherence to instructions on diaphragmatic breathing	No	App assumes to be equally as effective as in-person alternative without clinical evidence.	No	Less expensive at 1600 and 1500 users for military and civilians, resp.	US	Assumed	Yes	Yes	Yes	Yes	Costs	N/A	Costs	No	No	Yes, one way, probabilistic	Yes	No	No

Armstrong K, at al. (2014)	To assess the replacement of conventional, in-person postoperative follow-up care with mobile app follow-up care following ambulatory breast reconstruction in post-mastectomy breast cancer patients.	Bring-your-own-device (BYOD) mobile app to support postoperative care in breast reconstruction patients. App provides patients with validated quality of recovery questionnaires, surgical site photo submissions asynchronously- for health professionals to detect postoperative complications and eliminate in-person follow-up care.	Low-risk postoperative ambulatory patients	Usual care (in-person follow-up visit)	Societal perspective; Healthcare system perspective	30 days	Cost-effectiveness analysis	Successful surgical outcome (defined as a surgical patient not requiring medical or surgical intervention related to the original surgery within the first 30-days postoperative)	Incremental net benefit (INB)	All micro-cost data based on 2013/2014 estimates; clinic costs per patient from WCH; provided and physician fees from Ontario Health Insurance Plan (OHIP) billing codes	No, None. Effectiveness measure chosen based on literature.	1000 (hypothetical)	None	Follow-up feedback from patient (not well defined).	None	Yes (limited e-assessment billing codes, none for surgical e-assessments; lack of comparative effectiveness data; ICER not reportable)	No	App is cost-effective	Ontario, Canada	Yes	Yes	Yes	Maybe	Yes	Yes, micro-costing approach	Yes	Yes	Yes	Yes	Yes (scenario, two-way and probabilistic)	Yes	No	Yes

Ryan D, et al. (2012)	To determine whether mobile phone-based monitoring improves asthma control compared with standard paper-based monitoring strategies.	t+ Asthma application that enables twice daily transmission of symptoms, drug use, and peak flow; patients prompted to agreed action plan. Incursion into the red or amber zones triggered contact by an asthma nurse on the next working day.	Adolescents and adults with poorly controlled asthma	Standard paper-based monitoring strategies	Healthcare system	6 months	Cost-effectiveness analysis	Primary: Changes in scores on asthma control' self-efficacy; Secondary: mini-asthma quality of life, adverse occurrences, prescriptions of asthma drugs, modified patient enablement, engagement with process.	Asthma control' self-efficacy; mini-asthma quality of life, adverse occurrences, prescriptions of asthma drugs, modified patient enablement, engagement with process.	Asthma control questionnaire (ACQ), Knowledge, attitude, and self efficacy asthma questionnaire (KASE-AQ), Mini-asthma quality of life questionnaire (mini-AQLQ), Adverse occurrences obtained from the practice records, Prescriptions of asthma drugs recorded in the patients' healthcare record, Modified patient enablement instrument (mPEI), Engagement with process (proportion of patients defaulting from clinical follow-up)	Yes, RCT	288	age, sex, ethnicity, history of asthma, current smoking status, presence of comorbidity, FEV	Self-monitoring, self-efficacy	Self-monitoring, self-efficacy	Yes; small attrition rate, assumes non-responders did not improve, requirement for a compatible phone and network,	Yes	No significant difference, intervention more expensive (not cost-effective)	UK	Yes, or borrowed	Yes	Yes	Yes	Yes (not full description)	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes

Fawsitt CG, et al. (2017)	To assess the feasibility of an integrated mobile application versus a stand-alone app or telephone helpline at surveying surgical site infections.	Integrated mobile application for post-discharge monitoring after c-section discharge.	Women who deliver by c-section	Stand-alone app (usual care) and telephone helpline	Healthcare sector	6 weeks	cost-benefit analysis	WTP	Net benefit	Cost estimates derived from primary data collection, data from CUMH to estimate the expected capacity use over a six-week period.	No	380	None	Compliance with SSI prevention advice	None	Interchange terms CEA versus CBA, valuing costs and benefits	No	Cost-beneficial	Ireland	Yes	Yes	yes	yes	Yes	Yes	No	No	Yes	No	yes, probabilistic	no	no	no

B: PRISMA 2009 Checklist for Reporting Systematic Reviews and Meta-analysis

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

C: Key Recommendations from The Second Panel on Cost-Effectiveness in Health and Medicine

Recommendation 1: Reference Cases and Perspectives

All studies should report a reference case analysis based on a health care sector perspective and another reference case analysis based on a societal perspective. The reference cases are defined by recommendations for components to consider for evaluation, methods to use, and elements for reporting. It is recommended that reference case analyses measure health effects in terms of

QALYs. Standardizing methods and components within a perspective is intended to enhance consistency and comparability across studies.

Recommendation 2: Health Care Sector Reference Case

Results of the health care sector reference case analysis should be summarized in the conventional form as an incremental cost-effectiveness ratio. Net monetary benefit or net health benefit may also be reported, and a range of cost-effectiveness thresholds should be considered. In addition, the health care sector perspective should include formal health care sector (medical) costs reimbursed by third-party payers or paid out-of-pocket by patients. Both types of medical costs include current and future costs both related and unrelated to the condition under consideration.

Recommendation 3: Societal Reference Case

Recommendation 3A: Inclusion of an Impact Inventory

Evaluation of the broader effects of interventions designed to improve health is strongly recommended. The societal reference case analysis should include medical costs (current and future, related and unrelated) borne by third-party payers and paid out-of-pocket by patients, time costs of patients in seeking and receiving care, time costs of informal (unpaid) caregivers, transportation costs, effects on future productivity and consumption, and other costs and effects outside the health care sector. To make this evaluation more explicit and transparent, inclusion of an “impact inventory” that lists the health and non-health effects of an intervention should be considered in a societal reference case analysis. The main purpose of the impact inventory is to ensure that all consequences, including those outside the formal health care sector, are considered regularly and comprehensively, which has generally not been the case to date.

Recommendation 3B: Quantifying and Valuing Non-health Components in the Impact Inventory

Analysts should attempt to quantify and value non-health consequences in the impact inventory unless those consequences are likely to have a negligible effect on the result of the analysis.

Recommendation 3C: Summary and Disaggregated Measures

It would be helpful to inform decision makers through the quantification and valuation of all health and non-health effects of interventions, and to summarize those effects in a single quantitative measure, such as an incremental cost-effectiveness ratio, net monetary benefit, or net health benefit. However, there are no widely agreed on methods for quantifying and valuing some of these broader effects in cost-effectiveness analyses. Analysts should present the items listed in the impact inventory in the form of disaggregated consequences across different sectors. It is also recommended that analysts use 1 or more summary measures, such as an incremental cost-effectiveness ratio, net monetary benefit, or net health benefit, that include some or all the items listed in the impact inventory. Analysts should clearly identify which items are included and how they are measured and valued, and provide a rationale for their methodological decisions.

Recommendation 4: Reporting the Reference Cases and Other Perspectives

Recommendation 4A: Stating the Perspective

Analysts should clearly state the perspective of every analysis reported.

Recommendation 4B: Presenting other Perspectives

When specific decision makers have been identified, such as a public or private payer, analysts may want to present results from that decision maker’s perspective in addition to the 2 reference case perspectives. In these cases, analysts should indicate who the primary decision makers were whose deliberations are intended to be informed by the analysis.

Recommendation 4C: Importance of Transparency and Sensitivity Analysis

The items included in a cost-effectiveness analysis and the way they are valued involve numerous choices. Analysts should be transparent about how they have conducted the analyses, and convey how the results change with alternative assumptions. Sensitivity analysis should describe the assumptions to which the results for different perspectives are sensitive.

Recommendation 5: Designing a Cost-Effectiveness Analysis

1. Cost-effectiveness analysis (CEA), return-on-investment analysis, cost-minimization analysis, cost-consequence analysis, and cost-benefit analysis are alternative forms of analysis. The use of one does not preclude the use of the others.

2. We recommend that all studies report a Reference Case analysis based on a healthcare sector perspective and a Reference Case analysis based on a societal perspective. The Reference Cases include recommended methods, elements for reporting, and recommendations concerning what to include in the healthcare sector and societal perspectives. Inclusion of standardized components and standardization of methods within a perspective are intended to enhance consistency and comparability across studies.

3. We recommend that the results of the healthcare sector Reference Case be summarized in the conventional form, as an incremental cost-effectiveness ratio (ICER). The net monetary benefit (NMB) or net health benefit (NHB) may also be reported, and a range of cost-effectiveness thresholds should be considered. We recommend that the healthcare sector perspective analysis includes all formal healthcare (medical) costs—current and future, and related and unrelated—regardless of who bears the costs.

4. The Second Panel strongly supports evaluation of the broader impacts of health interventions. Thus, we recommend that analysts include a societal perspective Reference Case analysis. The societal perspective is essential whenever interventions are likely to have important effects on sectors of the economy outside of healthcare, and when there is a need or desire to understand all costs and effects regardless of to whom they accrue. The societal perspective includes medical costs (current and future, related and unrelated) borne by third-party payers or paid for out-of-pocket by patients, time costs of patients in seeking and receiving care, time costs of informal caregivers, transportation costs, effects on future productivity and consumption in added years of life, and other costs and effects outside the healthcare sector. The Second Panel agrees that it would be helpful to inform decision makers through the quantification and valuation of all health and non-health consequences of interventions, and to summarize those consequences in a single quantitative measure. However, there are no widely agreed upon methods for quantifying and valuing some of these broader impacts in CEA. We therefore recommend that analysts present the items listed in the Impact Inventory in the form of disaggregated consequences across different sectors. We also recommend that analysts use one or more summary measures, such as an ICER, NMB, or NHB, that include some or all the items listed in the Impact Inventory. Analysts should clearly identify which items are included and how they are measured and provide a rationale for their methodological decisions.

5. To help make the evaluation more explicit and transparent, we recommend inclusion of an Impact Inventory in every societal Reference Case analysis, even if the analyst does not quantify or value effects beyond the healthcare sector. The Impact Inventory lists the health and non-health impacts of an intervention that the analyst should consider in a societal Reference Case analysis.

6. All aspects of the interventions that may affect their cost or effectiveness should be defined for the analysis. These will include the target population and features such as the specific technologies used, the type of personnel delivering the intervention, the site of delivery, whether the service is “bundled” with other services, the frequency of the intervention, and its timing.

7. The scope of a study should be defined broadly enough to encompass the full range of groups of people affected by the intervention and all-important consequences. For example, if an intervention affects both mother and child, the analyst should include the health impacts and costs to each.

8. When there is heterogeneity in estimates of cost-effectiveness between different subpopulations, the analyst should present these differences as subgroup analyses. The analyst should be aware that subgroup analyses may raise ethical issues and questions of equity. The Second Panel provides a framework for consideration of ethical questions that may be important to the analyst, decision makers, and stakeholders.

9. Both Reference Case analyses should consider the full range of available and feasible options, including existing practice (the “status quo”) and a do-nothing option, as appropriate.

10. Options being compared may include more than specific interventions, extending, for example, to therapeutic strategies, alternative treatment starting and/or stopping rules, joint or sequential diagnostic test strategies, alternative levels of intensity of an intervention, and public health interventions. In all cases, these should be specified as mutually exclusive options and be compared using incremental CEA.

11. Costs and outcomes that have little impact on the estimate of cost-effectiveness or the likelihood of an intervention being cost-effective, and therefore little impact on the relevant decisions, can reasonably be excluded from an analysis. Their magnitude should be indicated in the Impact Inventory, clarifying the decision to exclude them.

12. The time horizon adopted in a CEA should be long enough to capture all differences between options in relevant costs and effects. 13. We recommend that analysts develop a written protocol for the design and conduct of the CEA. The protocol should describe, at a minimum, the objectives of the study, including the research questions; the type of economic evaluation to be conducted; the perspective of the analysis; the intervention and comparators; the population under consideration; the time horizon for the study; sources of data; a list of key assumptions; and an analysis plan, including the outcomes the analysis will assess. The analysts should update the protocol as the study progresses, and note any changes from the original protocol.